

What is DSA?

DSA stands for Data Structure and Algorithms.

It is a combination of two major components

1. Data Structure
2. Algorithms

So, let's understand these two components first before moving ahead.

What is Data Structure?

Data structure is the way of organizing and storing data in a computer.

Data Structure is like arranging your room efficiently so you can find things easily when it's needed the most.

Key features of Data Structure

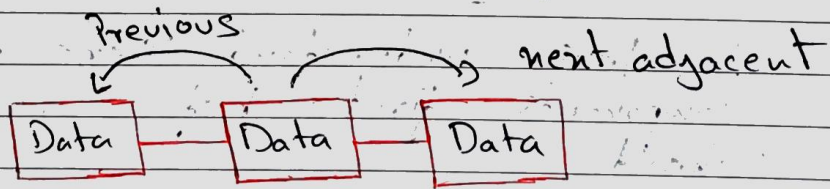
1. Efficiency
2. Flexibility
3. Reusability

Types of Data Structure

Data structure can be classified into two broader categories:

1. Linear Data Structure

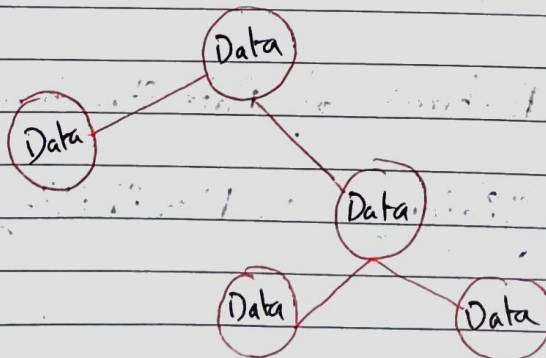
It is a way of arranging data sequentially or linearly, where each element is attached to its previous and next adjacent elements.

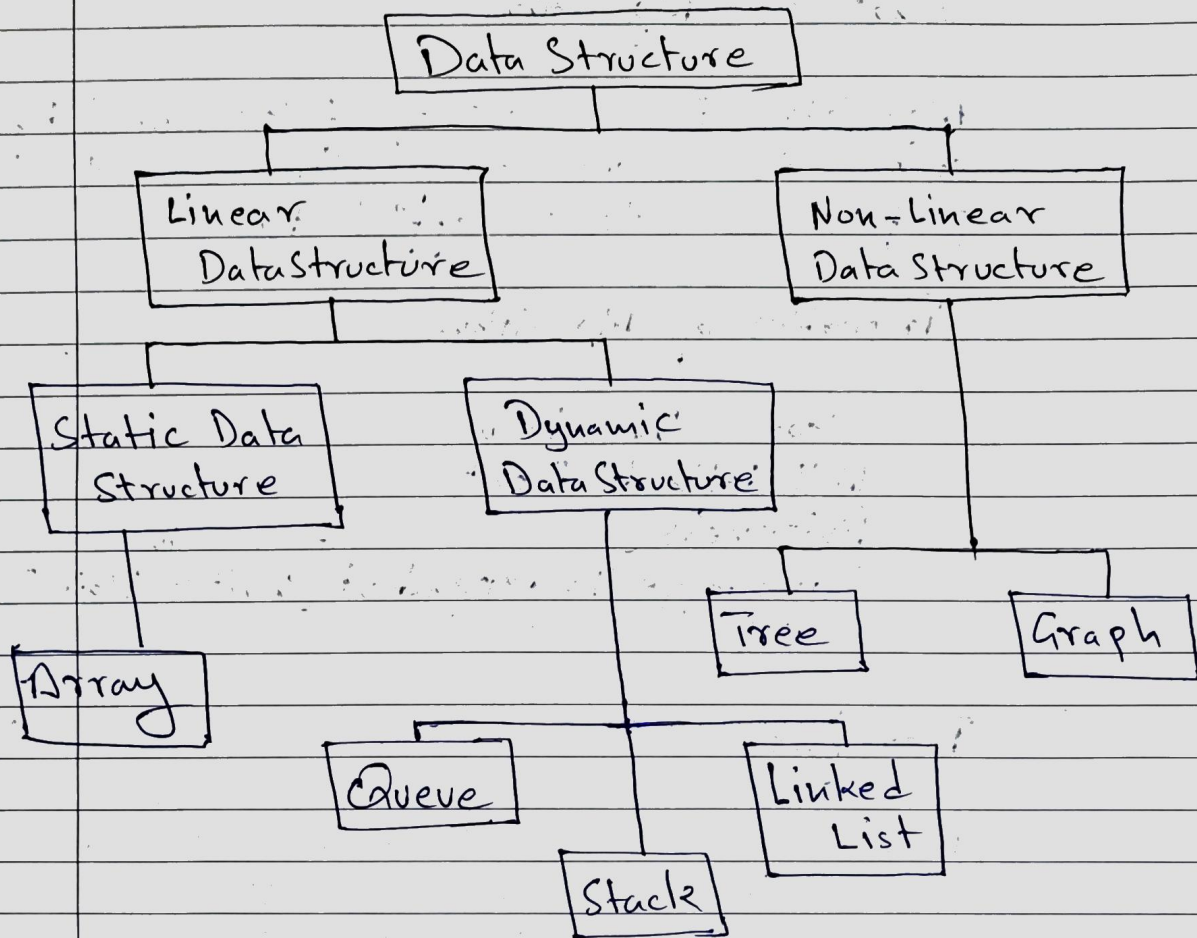


Example: Array, Stack, Linked List

2. Non-Linear Data Structure

It is a type of data structure where elements are not placed linearly, usually placed in form of trees and graphs.





Static : A data structure whose size is fixed before use.

Dynamic : A data structure whose size can change during runtime.

What is Algorithm

Algorithm is a set of finite, well define steps to solve a problem or to perform a task efficiently

Properties of Algorithm

Clear & Unambiguous

Well defined input

finiteness

Language Independent [Written Plain Text]

Therefore,

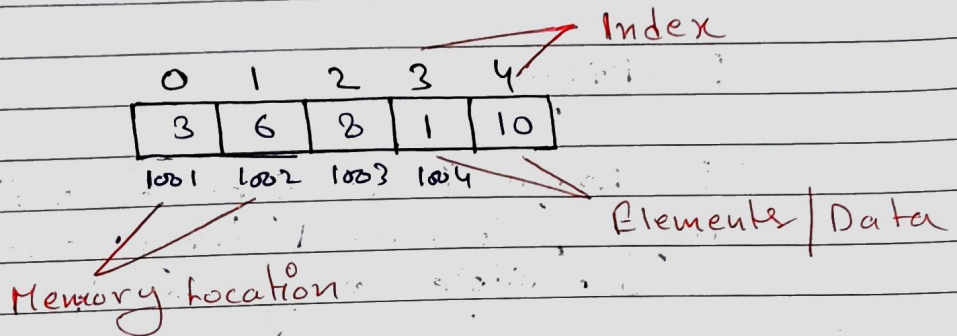
Data Structure and Algorithm (DSA) is the way of organizing data efficiently and solving problem efficiently.

It leads to resource optimization

Time / Memory

Array [Linear Data Structure]

Array is the collection of similar items stored at contiguous memory locations



Benifits of Array

- Stores multiple value together
- Easy to access data
- faster Searching and Processing

Syntax to declare array

In Java

1. `int[] arr = new int[5]` Size of Array
 (→ array declaration + Memory Allocation)
2. `int[] arr = {10, 20, 30, 40}`
 (→ array declaration + Initialization)

Multidimensional Array

Multidimensional array are Arrays of Arrays, Each elements of a M-D array itself is an array itself.

2D - Array

2D - Array is a type of multidimensional array, where, 1 array consist of multiple arrays as its elements.

It forms a matrix.

Representation of 2D-Array

	0	1	2	
0				→ Row 1
1				→ Row 2
2				→ Row 3
3				→ Row 4
	↓	↓	↓	
	col1	col2	col3	

In this matrix each row is a individual matrix array, stored as a element of 1 single array.

Let suppose there is a 2D array consist of 4 array elements, size of each array element is also 4

2D-Array = [Array1, Array2, Array3, Array4]

[1, 2, 3, 4]

[5, 6, 7, 8]

[9, 10, 11, 12]

Now let's visually understand it [13, 14, 15, 16]

0	1	2	3	4	} Array 1
1	5	6	7	8	} Array 2
2	9	10	11	12	} Array 3
3	13	14	15	16	} Array 4

In this box/Matrix each row is a individual array and each row is marked by the index of main array. The 2D-Array

Now see the Matrix, index 0 consist of Array 1

Index 1 consist of Array 2
" 2 " " Array 3
" 3 " " Array 4

Syntax to declare 2D Array in Java

1. Declare + Initialize

```
int[][] arr = new int[3][4];
```

Data type

Rows

Columns

(no. of sub arrays)

(length of sub arrays)

0	0	0	0
0	0	0	0
0	0	0	0

2. Declare & assign values

```
int[][] arr = {
    { 1, 2, 3 },
    { 4, 5, 6 },
    { 7, 8, 9 }
};
```

1	2	3
4	5	6
7	8	9

Traversal In 2D Array

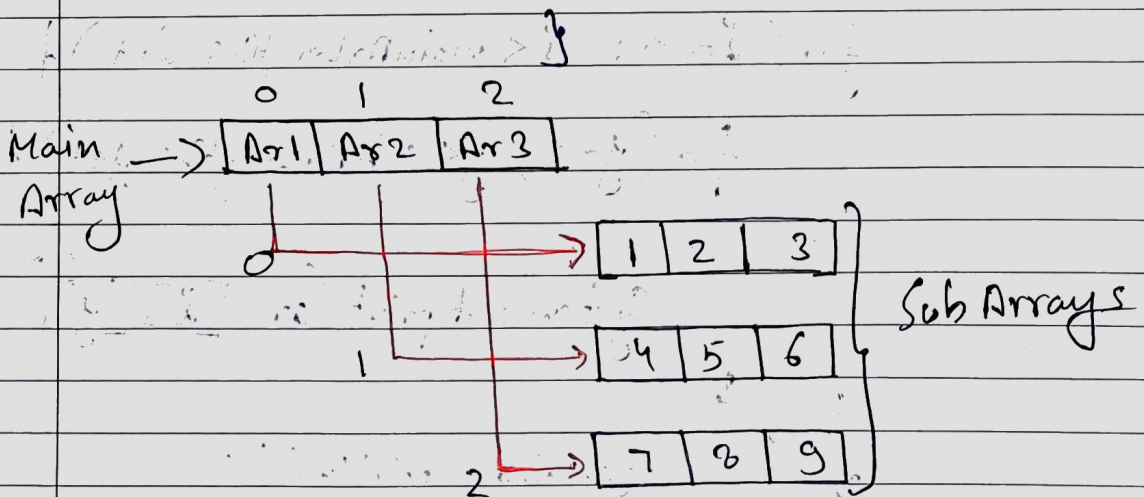
Traversal in 2D Array is done using 2 nested loops, in which

Outer loop → Controls the main array index

Inner loop → Controls the sub array index

Example:

```
int [][ ] arr = {
    { 1, 2, 3 },
    { 4, 5, 6 },
    { 7, 8, 9 }
}
```



So,

length of Main Array

```
int mainArLength = main arr.length
                  = 3
```

length of Sub array

```
int subArLength = arr[0].length
                 = 3
```

* take refrence of a diagram

Now, traversal

arr[0][1]

selecting sub array
on index 0

selecting index 1 of sub array

```
for(i=0; i < mainArLength; i++) {
```

```
    for(y=0; y < subArLength; y++) {
```

```
        System.out.print(arr[i][y]);
```

```
    }
```

```
System.out.println();
```

```
}
```